Aryan Mousavi

- **Email:** nirex.0@gmail.com
- **LinkedIn:** https://linkedin.com/in/aryan-mousavi
- **GitHub:** https://github.com/nirex0
- **Portfolio:** https://nirex.dev

Summary

Highly motivated and adaptable T-shaped Senior Software Developer with over 7 years of professional experience and a profound specialization in Deep Learning. Possesses a proven track record of rapidly mastering new technologies, underpinned by a strong mathematical and theoretical understanding of fundamental machine learning algorithms and a commitment to continuous study of emerging research. Combines 12 years of hands-on coding experience with a rigorous, 8-year self-study of deep learning, currently being formalized through a Bachelor's degree in Computer Science.

Technical Skills

- **Programming Languages:**
  - **Proficient:** C++, C#, Python, Dart
  - **Familiar with:** Java, JavaScript, SQL, PHP
- **Deep Learning & Machine Learning:**
  - **Core Concepts:** Feedforward Networks, Backpropagation, Gradient Descent, Optimization Methods, Loss Function Design, Double Descent phenomenon, Model Evaluation Metrics (Accuracy, Precision/Recall, F1, mAP). Grounded in a deep mathematical understanding of the underlying principles.
  - **Architectures:** Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), U-Nets, Transformers, Diffusion Models.
  - **Frameworks/Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, Transformers, Diffusers, OpenCV, Pillow
  - **ML Databases:** Vector Databases (Weaviate)
- **Developer Tools & Platforms:**
  - **Operating Systems:** Linux
  - **Simulation & Game Engines:** Unreal Engine 5
  - **General:** [e.g., Git, Docker, CI/CD]
- **Mathematical Foundations:**
  - Linear Algebra, Calculus, Probability & Statistics (Could use more practice)

Professional Experience

Senior Software Developer | Pasargad Flight Pioneers | Tehran, Iran | 2018 – 2025

- Delivered multiple revenue-generating projects for the company leveraging a diverse tech stack including C#, C++, Python, and Dart.
- **Example:** Unreal Engine Local Simulation for AI Training.
- **Example:** Delivered a flutter app that the company both used internally and sold to generate revenue.

Personal & Deep Learning Projects

Independent Deep Learning Researcher & Developer | 2016 - Present

- Implemented the Sophia Optimizer from scratch to deconstruct and understand its novel Hessian-free optimization mechanism (implementation available on GitHub).
- Implemented the Multi-Head Attention layer (both self- and cross-attention) from scratch to master the core component of the Transformer architecture.
- Architected and deployed a full-stack Virtual Try-on AI service from scratch on a self-hosted Linux server, featuring a Flutter mobile UI, a PHP backend, and a Python AI backend for processing requests.
- Designed and implemented a content-based Retrieval-Augmented Generation (RAG) system, utilizing a Weaviate vector database for efficient semantic search and retrieval.
- Developed a high-fidelity 3D simulation environment in Unreal Engine 5 for training and validating AI models, enabling the generation of synthetic data for robust model testing.
- Actively engages in ongoing research, exploring advanced topics such as using the Hessian matrix to analyze the curvature of the loss landscape for optimization stability.

Education

Bachelor of Science in Computer Science (In Progress)

- Azad University of Tehran
- **Expected Graduation:** 2026